2014 F=ma Exam: Problem 18

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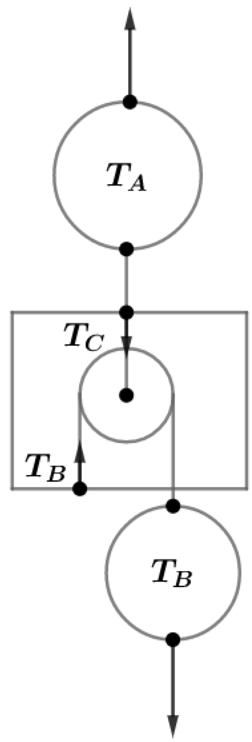

The tension from the two strings must balance the weight of the pulley. We have

$$T_C - 2T_B = M_p g$$

The tension from the three strings must balance the weight of the box:

$$T_A + T_B - T_C = M_b g$$

Adding these two equations, we obtain

$$T_A - T_B = (M_p + M_b)g$$

We could have directly obtained this by considering the external forces on the pulley-box system. Note the difference in tensions between T_A and T_B is constant. Thus,

$$T_A - T_B = 30\,\mathrm{N} - 20\,\mathrm{N} = 10\,\mathrm{N}$$

If T_B increases to $30\,\mathrm{N}$, then $T_A = T_B + 10\,\mathrm{N} = 40\,\mathrm{N}$ so the answer is $\boxed{\mathrm{B}}$.